

ANNA UNIVERSITY - MIT CAMPUS

Department of Computer Science and Engineering (CSE)



ULAVU

Direct Farmer–Retailer Produce Connection Platform

PROJECT DOCUMENTATION & TECHNICAL REPORT

CANDIDATE INFORMATION

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PROJECT RELEASE VERSION

Version 1.0 (Core Prototype)

Status: Verified & Operational

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1. Project Overview

ULAVU is a simple agricultural connectivity platform designed to connect farmers who have produce available with commercial retailers who are looking to procure it.

The platform focuses on one straightforward, essential interaction:

Core Interaction:

Farmer lists produce → Retailer discovers produce → Retailer contacts farmer directly.

Rather than attempting to become an over-engineered e-commerce portal with bidding, escrow payments, and rigid market restrictions, ULAVU is designed around **simplicity and accessibility**. It provides an immediate digital connection between rural cultivators and nearby commercial buyers with zero unnecessary friction.

2. Problem Statement

Agricultural produce distribution in regional markets faces several practical communication bottlenecks:

- **Difficulty in Reaching Buyers:** Farmers often struggle to discover and reach commercial buyers, supermarkets, and greengrocers beyond their immediate village gate or local commission brokers.
- **Fragmented Produce Availability:** Retailers and procurement managers lack real-time visibility into what neighboring farmers are harvesting on any given day. Sourcing often depends on informal word-of-mouth or costly intermediary networks.
- **Communication Inefficiency:** Information about available crop volumes is scattered across unorganized channels, creating localized gluts in one area while neighboring towns face procurement shortages.
- **Technological Alienation:** Existing agricultural apps often fail because they require lengthy onboarding, tax IDs, quality certificates, and multi-step pricing negotiations. Farmers are hesitant to adopt tools that require extensive data entry.
- **Need for a Simple Connection Layer:** Cultivators need a lightweight tool that makes their harvest discoverable without demanding excessive administrative effort.

3. Proposed Solution

ULAVU solves this challenge through a deliberately simple, two-sided workflow:

Farmer

The farmer creates a minimal profile containing only:

- **Name**
- **Mobile Number**
- **Location** (Village / Town / District)

To post available harvest produce, the farmer enters only:

- **Produce Name** (e.g., Tomato, Onion, Watermelon, Banana, Brinjal)
- **Quantity** (Numeric value)
- **Unit** (Strictly restricted to **kg** or **bag**)
- **Location** (Where the crop is available)

No pricing, budgets, or harvest expiration dates are required. Pricing is discussed and agreed upon naturally during direct conversation.

Retailer

The commercial buyer accesses the live produce feed:

- Browses available produce batches posted by regional farmers.
- Each listing clearly displays: **Produce Name**, **Quantity**, **Location**, **Farmer Name**, and **Contact Number**.
- Retailer clicks **Call Farmer** to immediately connect with the grower over phone and finalize the purchase.

4. Benefits

Benefits to Farmers

- **Quick Produce Listing:** Farmers can publish an available harvest batch in under 15 seconds.
- **Minimal Data Entry:** Only requires crop name, quantity in kg/bag, and location.
- **Direct Buyer Access:** Connects directly with commercial retailers and supermarkets across the region.

- **Zero Pricing Pressure:** Avoids forced digital price bidding; terms are finalized via natural phone conversation.

Benefits to Retailers

- **Effortless Discovery:** Immediate visibility into real farm supply currently available in surrounding districts.
- **Clear Volume Metrics:** Instant clarity on whether stock is available in kilograms or bags.
- **Location Transparency:** Displays exact towns and villages to evaluate sourcing distances.
- **Direct Contact:** Eliminates middleman delays with a single click to call the farmer.

Social Impact

- **Reduced Post-Harvest Loss:** Faster discovery connects perishable vegetables and fruits with demand before spoilage occurs.
- **Digital Inclusion:** Proves that rural technology adoption increases when complexity is eliminated.
- **Fairer Agricultural Trade:** Encourages transparent, direct relationships between rural food producers and urban retailers.

5. Uniqueness & Philosophy

Core Philosophy:

ULAVU is designed around simplicity. Instead of forcing farmers to enter extensive information into a complicated marketplace, it asks only for the information required to make their available produce discoverable.

Most agricultural software fails because it assumes rural cultivators want to operate like corporate enterprise traders. ULAVU focuses on what farmers currently have harvested today, rather than complex bidding sheets. Software is used purely for discovery, while negotiation happens over direct phone calls.

Additionally, the modular architecture provides a clean foundation for future voice and SMS integration, ensuring that farmers with limited smartphone literacy will eventually participate through simple keypad phones.

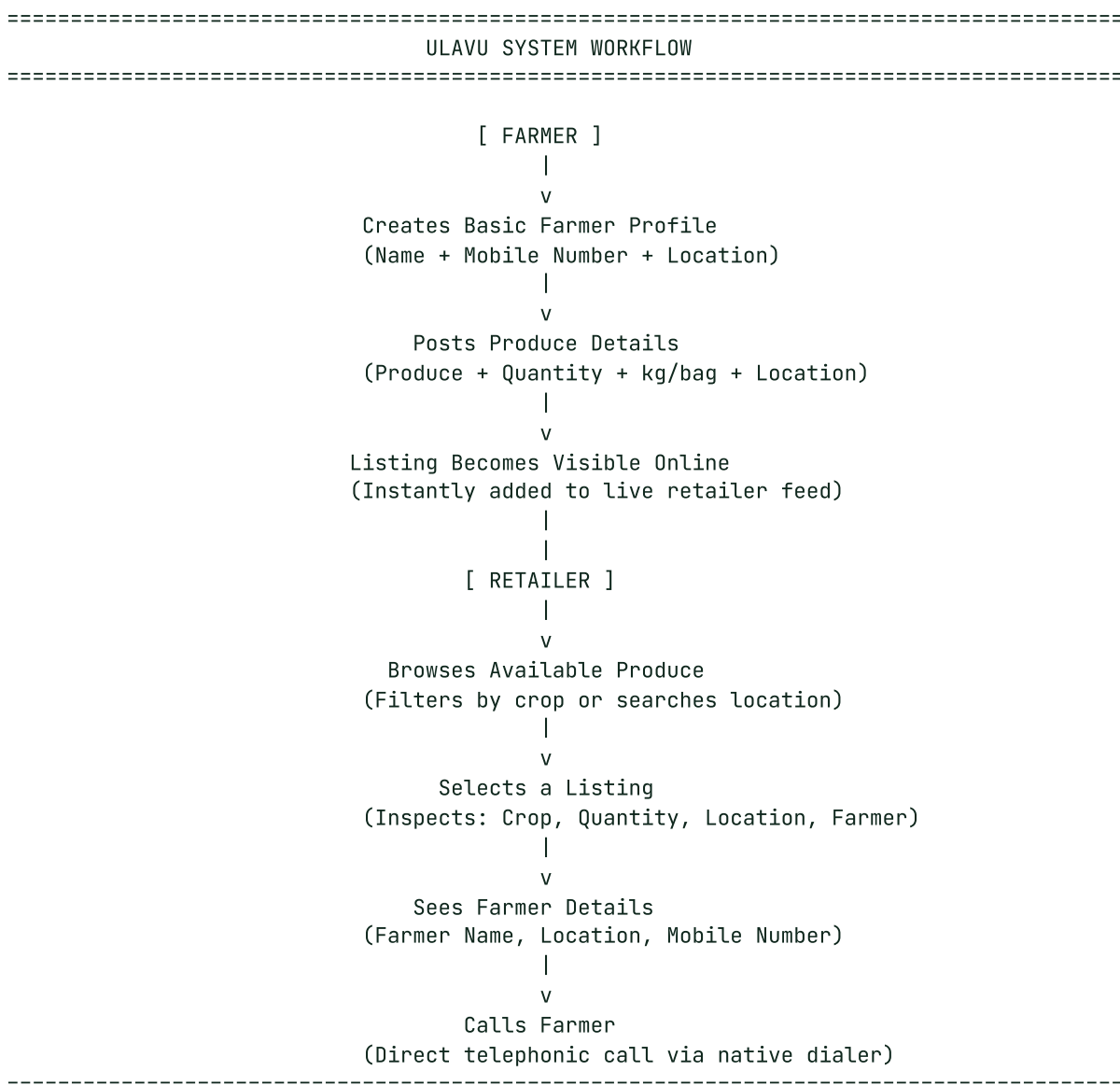
6. Six-Module Roadmap

ULAVU is structured conceptually into six modules. The current prototype fully delivers the core operational tier (Modules 1–3), while Modules 4–6 represent planned future extensions:

Module	Description	Current Status
Module 1	Farmer Profile & Produce posting in kg/bag	IMPLEMENTED
Module 2	Retailer Discovery feed of available farm produce	IMPLEMENTED
Module 3	Direct Farmer–Retailer telephonic connection	IMPLEMENTED
Module 4	Trust & Verification, ratings, and transparency	FUTURE SCOPE
Module 5	Analytics & Market Intelligence and supply insights	FUTURE SCOPE
Module 6	Voice & SMS Accessibility for non-smartphone users	FUTURE SCOPE

7. Workflow

The complete operational workflow is visually clear, direct, and free of administrative bottlenecks:



8. Technology Stack

The ULAVU prototype is built entirely with verified, modern web technologies genuinely present in the codebase:

Technology	Genuine Implementation	Purpose in ULAVU
React 19	Frontend library	Component-based reactive user interface
Vite 8	Build tool & dev server	High-speed bundling and development environment
Vanilla CSS	CSS3 design system	Custom agricultural styling, Flexbox, and Grid
Web Storage API	LocalStorage	Client-side data persistence across refreshes
RFC 3966 URI	tel: protocol	Direct farmer telephone calling via device dialer
Node.js / npm	Runtime & packaging	Local environment and dependency execution

9. Future Scope

The architecture cleanly accommodates three future modules without modifying the existing core:

- **Module 4 — Trust & Verification:** Farmer verification via Kisan credentials, commercial trade license checks for buyers, and community reputation ratings.
- **Module 5 — Analytics & Market Intelligence:** District-level crop availability heatmaps, seasonal production insights, and APMC Mandi benchmark price tracking.
- **Module 6 — Voice & SMS Accessibility:** An Interactive Voice Response (IVR) phone gateway where farmers can list crops and receive buyer inquiries via voice calls and SMS without needing a smartphone.

10. Developer Information

Name	Sujith B
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Description	<i>"I'm a curious and driven tech enthusiast who enjoys learning, building, and experimenting with technology. I'm especially interested in software development, cloud computing, Linux, and AI, and I like turning ideas into practical projects while continuously improving my skills."</i>

11. Conclusion

ULAVU focuses on solving one practical problem well: **helping farmers make their available produce visible to retailers and enabling direct communication between them.**

By avoiding complicated onboarding and eliminating unneeded pricing forms, ULAVU creates immediate, practical value for both rural producers and commercial buyers. Its modular six-tier design ensures that the current prototype provides a solid, extensible foundation for future trust, market intelligence, and voice/SMS accessibility capabilities.

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